Mathematical Statistics

Lecture 08: The Bernoulli Distribution

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1 Introduction to the Bernoulli Distribution

This lecture introduces the most fundamental discrete probability distribution: the **Bernoulli Distribution**. It models a single experiment with only two outcomes and serves as the essential building block for more complex statistical models.

Why This Matters for Math Students

The Bernoulli distribution provides the mathematical language for modeling binary, "yes/no" events. Mastering this concept is the first step toward understanding probabilistic machine learning models. It connects simple probability theory to powerful applications like predictive modeling (e.g., logistic regression) and statistical inference (e.g., A/B testing). It is the atom from which entire molecules of statistical theory are built.

2 The Bernoulli Trial: Concept and Examples

A Bernoulli distribution describes the outcome of a single experiment, a **Bernoulli trial**, which has the following key characteristics:

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- **Single Trial:** The experiment is performed only once.
- **Two Possible Outcomes:** The trial can only result in one of two outcomes, which we label for mathematical convenience:
  - **Success:** Assigned the numerical value **1**.
  - **Failure:** Assigned the numerical value **0**.

2.1 Framing an Experiment as a Bernoulli Trial

A crucial skill is recognizing how to frame scenarios as Bernoulli trials, even when the underlying experiment has more than two outcomes.

2.1.1 Example: Rolling a Single Die

An excellent example is rolling a standard six-sided die. While there are six possible outcomes {1, 2, 3, 4, 5, 6}, we can frame it as a Bernoulli trial by defining a specific event of interest.

- **Bernoulli Framing:** Let's define "success" as rolling a 6.
- **Success ($X = 1$):** The outcome is a 6. The probability is $P(X = 1) = 1/6$.
- **Failure ($X = 0$):** The outcome is not a 6, i.e., {1, 2, 3, 4, 5}. The probability is $P(X = 0) = 5/6$.

This creates a binary "yes/no" scenario—"Was the roll a 6?"—which is a perfect Bernoulli trial.

Table 1: Probability Distribution Table for the Die Roll Example

Random Variable (X)	**Outcome(s)**	**Probability $P(X)$**
0	{1, 2, 3, 4, 5}	$5/6 \approx 0.833$
1	{6}	$1/6 \approx 0.167$

3 Mathematical Foundations

Core Formulas of the Bernoulli Distribution

A Bernoulli-distributed random variable, denoted $X \sim \text{Bern}(p)$, is fully defined by a single parameter, p.

- **Parameter (p):** The probability of success, where $0 \leq p \leq 1$.
- Probability of success: $P(X = 1) = p$
- Probability of failure: $P(X = 0) = 1 - p$

1. Probability Mass Function (PMF)

The PMF gives the probability of any outcome $x \in \{0, 1\}$ in a single, compact formula:

$$f(x) = P(X = x) = p^x (1 - p)^{1-x}$$

This formula works because if $x = 1$, the $(1 - p)$ term becomes $(1 - p)^0 = 1$, leaving $p^1 = p$. If $x = 0$, the p term becomes $p^0 = 1$, leaving $(1 - p)^1 = 1 - p$.

2. Expected Value (Mean)

The expected value, $E[X]$, is the long-term average outcome of the trial, calculated as $E[X] = \sum x \cdot P(X = x)$.

$$\begin{aligned} E[X] &= (0 \cdot P(X = 0)) + (1 \cdot P(X = 1)) \\ &= (0 \cdot (1 - p)) + (1 \cdot p) \\ &= 0 + p = p \end{aligned}$$

Thus, the expected value is simply the probability of success: $E[X] = p$.

3. Variance

The variance, $\text{Var}(X)$, measures the spread of the distribution, calculated using the formula $\text{Var}(X) = E[X^2] - (E[X])^2$.

1. **First, find** $E[X^2]$ (the expected value of X^2):

$$\begin{aligned} E[X^2] &= \sum x^2 \cdot P(X = x) \\ &= (0^2 \cdot P(X = 0)) + (1^2 \cdot P(X = 1)) \\ &= (0 \cdot (1 - p)) + (1 \cdot p) = p \end{aligned}$$

2. **Now, apply the variance formula**:

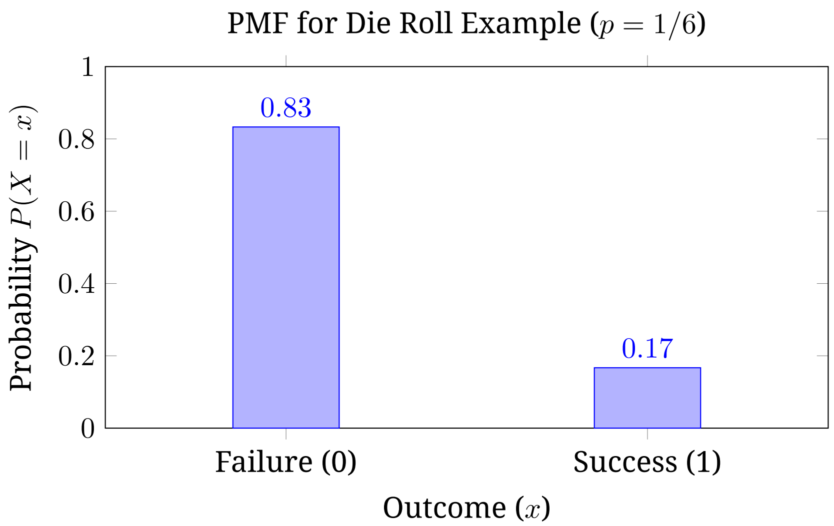
$$\begin{aligned} \text{Var}(X) &= E[X^2] - (E[X])^2 \\ &= p - (p)^2 = p - p^2 \end{aligned}$$

Factoring out p, the final formula for variance is **Var**$(X) = p(1 - p)$.

4 Visualizing the Bernoulli Distribution

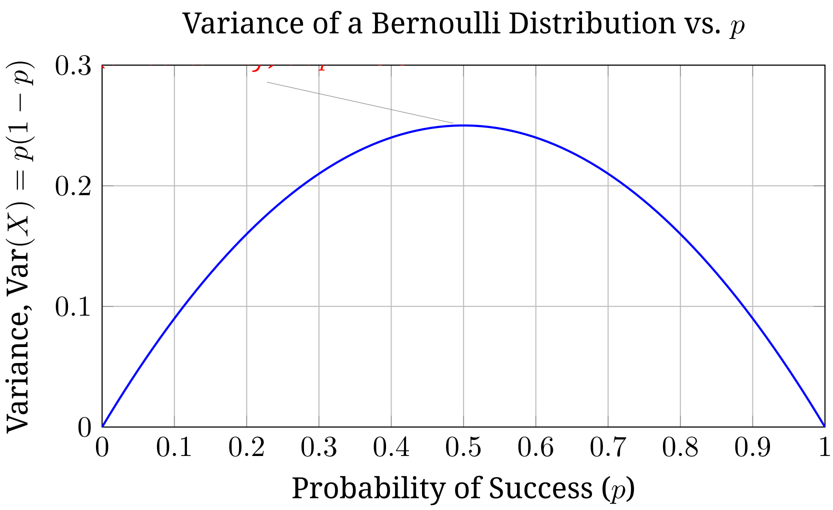
4.1 PMF Bar Chart for the Die Roll Example

A bar chart is the ideal way to visualize a discrete distribution. This chart clearly shows that failure is much more probable than success for our die roll example ($p = 1/6$).



4.2 Variance as a Function of p

The variance, $p(1 - p)$, reveals a key insight about uncertainty.



The variance is maximized when $p = 0.5$. This makes perfect intuitive sense: a fair coin ($p = 0.5$) has the **maximum uncertainty** about its outcome. If $p = 0$ or $p = 1$, the outcome is certain, and the variance is 0.

5 Applications in Data Science & Machine Learning

Application 1: A/B Testing

Imagine an e-commerce company testing if changing a "Buy Now" button from blue to green increases clicks.

- **Setup:** Show the blue button to Group A and the green button to Group B.
- **Bernoulli Trial:** For each user, the act of clicking (or not) is a Bernoulli trial where Click=1 (success) and No Click=0 (failure).
- **Modeling:** We model the outcomes as $X_A \sim \text{Bern}(p_A)$ and $X_B \sim \text{Bern}(p_B)$. The goal is to estimate p_A and p_B from the data to see if $p_B > p_A$. This simple probabilistic model allows us to make a data-driven business decision.

Application 2: Foundation of Logistic Regression

Logistic Regression is a fundamental classification algorithm that models the probability of a binary outcome.

- **The Link:** The outcome (e.g., Pass/Fail) is a Bernoulli variable. However, the probability of success, p, is not fixed; it **depends on the input features** (e.g., hours studied).
- **How it Works:** Logistic Regression uses a sigmoid function to map the input features to a value between 0 and 1. This output is the predicted probability p for the Bernoulli trial. For a student who studied 8 hours, the model might predict $p = 0.95$. Their outcome is thus modeled as $Y \sim \text{Bern}(0.95)$.

Other common applications include modeling customer churn, medical diagnoses, transaction fraud, and spam email detection.

Career Pathways & Future Scope

The Bernoulli distribution is a gateway to more advanced concepts. It is the single-trial version of the **Binomial Distribution**, which models the number of successes in many trials. In models like **Naive Bayes**, Bernoulli distributions are used to handle binary features (e.g., "does the email contain the word 'free'?"). A solid grasp of this concept is essential for careers in data science, quantitative analysis, and biostatistics.

6 Practice Problems

1. A factory produces light bulbs, and it is known that 5% of them are defective. A single bulb is randomly selected and tested. Let $X = 1$ if the bulb is defective and $X = 0$ otherwise.

   (a) What is the value of the parameter p for this Bernoulli trial?

   (b) What is the expected value, $E[X]$?

   (c) What is the variance, $\text{Var}(X)$?

2. You are playing a game where you win if you draw an Ace from a standard 52-card deck.
   (a) Define the random variable X, the "success" and "failure" outcomes, and the parameter p.
   (b) What is the probability of failure?
3. A Bernoulli random variable X has an expected value of 0.3.
   (a) What is $P(X = 1)$?
   (b) What is $P(X = 0)$?
   (c) What is the variance of X?

7 Solutions to Practice Problems

1. **Light Bulb Problem:**
   (a) Success is defined as the bulb being defective. The probability of this is 5%. So, $\boldsymbol{p = 0.05}$.
   (b) The expected value is $E[X] = p$. So, $\boldsymbol{E[X] = 0.05}$.
   (c) The variance is $\text{Var}(X) = p(1-p) = 0.05 \times (1-0.05) = 0.05 \times 0.95 = \mathbf{0.0475}$.

2. **Card Game Problem:**
   (a) The random variable X represents the outcome of the draw.
   - **Success ($X = 1$):** Drawing an Ace.
   - **Failure ($X = 0$):** Drawing any other card.

   There are 4 Aces in a 52-card deck. So, the parameter $\boldsymbol{p = \frac{4}{52} = \frac{1}{13}}$.
   (b) The probability of failure is $1 - p = 1 - \frac{1}{13} = \mathbf{\frac{12}{13}}$.

3. **Expected Value Problem:**
   (a) We know that $E[X] = p$, and $P(X = 1) = p$. Since $E[X] = 0.3$, then $\boldsymbol{P(X = 1) = 0.3}$.
   (b) $P(X = 0) = 1 - p = 1 - 0.3 = \mathbf{0.7}$.
   (c) The variance is $\text{Var}(X) = p(1-p) = 0.3 \times (1-0.3) = 0.3 \times 0.7 = \mathbf{0.21}$.